# Build guide 4: axes, electronics and spindle

DIY CNC epoxy-granite mill · October 2026

## 6. Axes

**Y (table).**

1. Bolt the Y rails (full 600 mm, centerd 20 mm behind the base split) to their risers (M4 x 16, about 4 N·m). Tighten the first rail against a straightedge, then set the second parallel with a dial indicator on a block of the first (within 0.01-0.02 mm over the travel).
2. Bolt the HM12-57 bracket (the fixed bearing and the motor mount in one, rear; its six holes) and the BF12 (front) to their tapped steel pads (M6 x 40, about 6-8 N·m) with the screw installed. Move the bearings, spacer and lock nut from the kit's BK12 into the bracket if it came without them. Align the screw parallel to the rails in both directions with a dial indicator on the nut. Shim under a support if needed; the screw should spin freely by hand along the whole length.
3. Mount the DSG16H nut housing under the table (shim to height), bolt the table to the four blocks and the housing, and check for binding at both ends of travel. Fit the 8 x 10 coupler (the length the bracket takes, L30-42) and the closed-loop motor on the bracket.

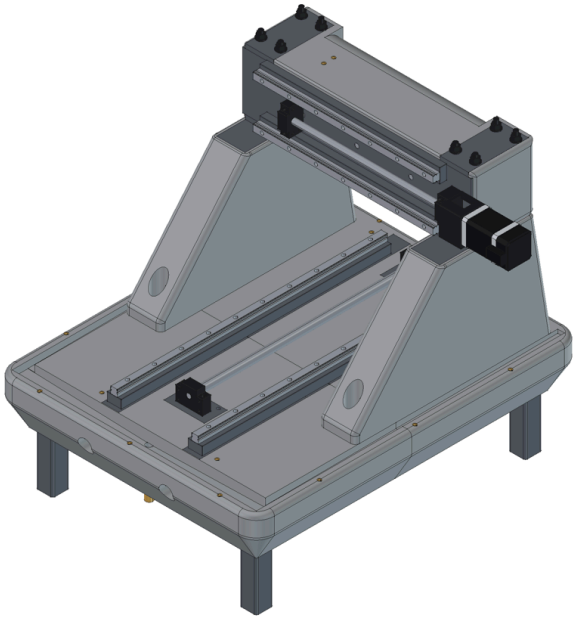
**X (beam).**

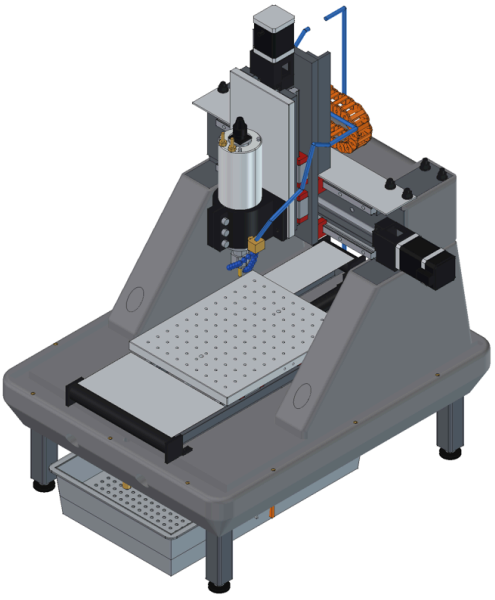
1. Rails onto the X risers as for Y (100 mm spacing).
2. BK10 and BF10 onto the beam's face bar, tapped M6: M6 x 40 (the casting is relieved behind each hole, so the screws don't bottom). Align and check as for Y.
3. The X carriage is 5/8 in steel: drill and tap it to the parts it carries (the four X blocks and the nut housing on the back, the four Z blocks and the Z BK10/BF10 on the front, M4 and M6). Use a drill press, spiral-point taps and cutting oil; mark from the parts themselves or from a printed layout. Drill the two 1-1/4 in angles (five 17/64 in holes each, from the steel templates), clamp each one up the back of the carriage at its outer edge, flush with the top, spot the carriage through the holes, tap it M6 and bolt them on (M6 x 16). Paint or oil the bare faces: the coolant reaches it.
4. Bolt the carriage to the four blocks and the nut housing; motor bracket, 8 x 8 coupler, closed-loop motor.

**Z.**

1. Z risers onto the Z plate: tap the risers M5 and screw them on from the back of the plate (no inserts needed in aluminum). Rails on the risers, and the four Z blocks on the front of the X carriage.
2. BK10 / BF10 and the Z screw on the carriage, nut housing on the Z plate, open-loop motor on top.
3. Spindle clamp on the Z plate, spindle in the clamp. Tram it later (section 9).

Lubricate every block before moving anything far: a grease syringe with a needle tip on the blocks' M4 nipples does until the grease gun (an add-later item).





## 7. Electronics and spindle

- **Controller:** FluidNC board with external drivers. CL57T drivers for X and Y (closed loop), DM542T for Z, 48 V supply trimmed to about 46 V (the CL57T maxes out at 48 V), 24 V logic.

- **Steps:** with drivers at 1600 pulses/rev: Y (SFU1605, 5 mm lead) 320 steps/mm; X and Z (SFU1204, 4 mm lead) 400 steps/mm.
- **Spindle:** VFD for 120 V now (limits it to about 1.5 kW), the kit's 220 V VFD later on 240 V. Use a dedicated 20 A circuit and shielded spindle cable, earthed at the VFD end. Set the VFD for the spindle's nameplate: 400 Hz, 220 V, 24,000 rpm.
- **Spindle cooling:** water pump in a 10-20 L bucket of distilled water with 10-20 % RV antifreeze, switched by a relay so it runs whenever the spindle runs. Never run the spindle dry.
- **Safety:** wire the E-stop to cut driver enable and the VFD run signal in hardware. Fit the homing switches, then the tool setter. None of the switches needs anything in the castings: Y goes on the Y motor bracket, where the table's rear edge (or a small printed dog on it) trips it; X and Z go on the X carriage, X tripped by the BK10 near the end of travel, Z by the Z plate.